

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

May 2012

EC 305 Microcontroller & Application

Date: 04/05/2012, Friday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Briefly give answers of the following questions: [10]

1. What are the advantages and disadvantages of using Harvard architecture in 8051?
2. What is the purpose of EA pin in 8051 microcontroller?
3. What is the function of SMOD bit in TCON SFR?
4. Give difference between PC and DPTR pointers.
5. Write instruction to select register bank 3.

Q - 2 (a) Describe and Draw Architecture of intel-51, 8-bit Microcontroller. Discuss ROM and RAM structures. [05]

(b) Explain operation of timer in mode 2. Write program to generate delay of 0.1 second using timer 0 in mode 2. Assume suitable crystal frequency. [05]

(c) Explain following Addressing Modes of 8031 Microcontroller [05]

1. MOV A, #20H
2. MOV A, 30H
3. MOV A, @R0
4. MOVX A, @DPTR
5. MOVC A, @A+DPTR

OR

Q - 2 (a) Differentiate following: [04]

1. PC vs DPTR
2. JMP vs CALL instruction

(b) Draw and explain 8051 connection to external RAM (8Kx8). Write a program to read 100 bytes of data from P1 and save the data in external RAM starting at 6750H Location. [07]

(c) What can be the maximum distance from the address of the jump instruction in case of SJMP, AJMP and LJMP? [04]

Q - 3 (a) Draw the interfacing diagram of external 16K EPROM and 4K RAM with the μ C 8051. Give address range of each memory type. [05]

(b) Write four different methods of instruction program to clear accumulator. [02]

(c) Write a program to find complement of accumulator without using CPL. [03]

OR

Q - 3 (a) Write a program to transmit message "CHARUSAT" serially at baud rate of 9600, 8-bit data and 1 stop bit continuously. [05]

(b) Explain serial communication mode 0 and mode 1 in 8051. Also write the format of SCON register. [05]

SECTION – II

Q - 4 (a) Describe the function of following instructions of 8051. Which flags are affected after execution of the instruction? [07]

- | | |
|---------|------------------|
| 1. SUBB | 5. CLR |
| 2. DEC | 6. DJNZ R0,label |
| 3. SWAP | 7. MOVX A,@DPTR |
| 4. DAA | |

Q - 5 (a) Draw interfacing diagram of 8051 with LCD. Write program to display message "EDUCATION CAMPUS CHANGA" on the LCD screen. [08]

(b) Calculate the Timer 0 reload value needed to achieve time delay of 40 msec. Oscillator frequency is 6 MHz. [06]

OR

Q - 5 (a) Write a program to generate 5 kHz pulse waveform of 50% duty cycle on pin P1.0 using timer 0. [08]

(b) If micro-controller is connected with crystal of frequency 11.0592 MHz, calculate the time required to execute the following instructions? [06]

1. RETI
2. MUL AB

Q - 6 (a) 16 keys are connected to port 2 using 4x4 key matrix. Write a program to identify the key pressed. Call a delay subroutine "DELAY" to generate debounce time of 20 msec. [10]

(b) Find the Iout for the following inputs. Assume Iref = 1.5 mA for DAC080. [04]

1. 55H
2. A3H

OR

Q - 6 (a) Draw interfacing diagram to interface push-button switch at port pin P3.0 and relay with transistor circuit at port pin P2.0. Write program such that when push-button key is pressed, relay should become ON. When push-button key is pressed again relay should become OFF (i.e. toggling of relay by push-button switch). [10]

(b) Draw circuit diagram to interface common cathode type 7-segment display with microcontroller 8051. [04]